

Yassine Abassi

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EDUCATION

Polytechnique Montréal

Bachelor's in Computer Engineering
Preparatory Program

Montreal, QC

Aug 2022 – May 2026

Aug 2021 – May 2022

Bayram 5 Pioneer High School

Baccalaureate in Mathematics, Summa Cum Laude

Tunis, Tunisia

Sep 2017 – June 2021

EXPERIENCE

Research Software Intern

FIU-Center for Translational Science

May 2026 – July 2026

Port St. Lucie, FL

- Built a **Python/RDKit** cheminformatics tool to automate combinatorial molecule enumeration from Excel data, converting chemical names to SMILES, assembling branch combinations onto a molecular scaffold, applying stereochemistry rules, and exporting a validated library of **10,000+** compounds for research use.

Software Developer

Hydro-Québec

Jan 2026 – May 2026

Montreal, QC

- Built a full-stack weather visualization platform using **Angular** and **ASP.NET Core**, featuring multi-model weather display with interactive geospatial mapping via **Leaflet** and real-time updates via **SignalR**.
- Implemented **GDAL** raster processing for cartographic rendering; designed **RESTful APIs** backed by an **SQLite** database via **EF Core**, implemented caching and persistence strategies to improve performance, containerized the application with **Docker**, and set up a **CI/CD** pipeline for automated testing and deployment.

Software QA Automation Intern

UpToTest

May 2024 – Sep 2024

Tunis, Tunisia

- Architected a **Cypress/Cucumber** BDD test framework using a **POM** design pattern, implementing **400+** end-to-end scenarios; integrated the suite into **GitLab CI** for automated regression, achieving **89%** feature coverage.
- Executed cross-browser test runs against a dedicated pre-production environment and configured **Mochawesome** HTML reports delivered alongside an Agile Scrum team every sprint providing full run visibility.

Private Tutor

Self-employed

Aug 2021 – Present

Montreal, QC

- Provided in-person tutoring in Mathematics, Physics, and Programming to **20+** college and university students, through personalized explanations, structured exercises, and exam-focused problem solving.

PROJECTS

3D Interactive Portfolio

[Live Demo](#) 📄

- Crafted a 3D portfolio in **React.js** and **Three.js** featuring an interactive room with custom camera rigging.
- Expanded the user experience with terminal emulation, procedural art, and embedded mini games; integrated a Gemini-powered chatbot assistant with **Upstash Redis**-backed rate limiting.

Corporate LinkedIn Translator

[GitHub](#) 📄

- Built and published a Chrome extension that uses Google Gemini AI to detect and translate LinkedIn corporate buzzwords into plain satirical language with inline tooltip annotations.
- Architected a full **Node.js/Express** backend API deployed on **Fly.io** with rate limiting, graceful error handling, and dynamic DOM injection to handle LinkedIn's React SPA structure.

WildGuard — AI Wildlife Monitoring System

[GitHub](#) 📄

- Built an AI-powered ecological surveillance platform in **Python** for anti-poaching and endangered species protection, with real-time detection updates via **Socket.IO** and automated SMS alerting through **Twilio**.
- Implemented a **YOLOv8** detection model capable of identifying **50+** species, integrated with a **Flask** backend.

Multi-Robot Exploration System

- Designed a coordinated multi-robot exploration system in **Python** for **AgileX Limo** robots using **ROS2** and **Gazebo** for simulation, implementing custom nodes for inter-robot communication and task distribution.
- Containerized software modules using **Docker** for reproducible deployment across physical and simulated environments.

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, Go, VHDL

Web Frameworks: Angular, React.js, Node.js, NestJS, Express.js, WebSockets, SignalR, .NET, Three.js

Data & APIs: PostgreSQL, MongoDB, SQLite, Firebase, Retrofit, EF Core, GDAL, Upstash Redis

DevOps & Tools: Git, GitHub Actions, GitLab CI, Docker, Jira, Fly.io, Cypress, Vivado

Embedded & Hardware: Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo